

Total No. of Questions : 10]

SEAT No. :

P3126

[5154]-692

[Total No. of Pages : 2

B.E.(Information Technology)
SOFTWARE MODELING AND DESIGN
(2012 Pattern) (Semester-I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw appropriate diagrams using UML 2.0 notations.*
- 4) *Use of non programmable electronic calculator is allowed.*

- Q1)** a) Elaborate the keywords class, association name, association end name and multiplicity with an example. [6]
- b) Elaborate extend, include with the context of use case diagram using an example. [4]

OR

- Q2)** a) Elaborate on association, aggregation and composition with an example. [6]
- b) Write a short note on representation of constraints using UML2.0 [4]
- Q3)** a) Define a state, transition, entry, exit, and do action in the context of state diagram. [6]
- b) Explain the expansion region in the context of activity diagram. [4]

OR

- Q4)** a) What is the relation between use cases and sequence diagram? Explain the keywords participants, time line, focus of control, synchronous message with respect to sequence diagram. [6]
- b) Elaborate composite state and concurrent state with an example. [4]
- Q5)** a) Describe batch transformation and continuous transformation. [8]

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- b) Write the purpose of deployment diagram. Draw & explain the following element of deployment diagram. [8]

- i) Node
- ii) Artifact
- iii) Node instance

OR

- Q6)** a) Explain layered architecture & partitions. [8]

- b) Draw component diagram for online shopping system. [8]

- Q7)** a) What is design pattern? Explain 4 essential elements of patterns. [8]

- b) Explain the Design pattern documentation. [8]

OR

- Q8)** a) Write a short note on [8]

- i) Observer design pattern.
- ii) State design pattern.

- b) Write the classification, motivation, class diagram and uses of adapter design pattern. [8]

- Q9)** a) Differentiates between Black box testing and white box testing. [8]

- b) Draw and explain V-model of testing. [10]

OR

- Q10)** a) Test Driven development: Explain in brief [8]

- b) Define software validation and software verification. [10]

Explain verification and validation concept by considering the following statements:-

- i) Are we building the product right?
- ii) Are we building the right product?

